Developer resources alternative OS overview

How to connect Sylvac Bluetooth® instruments with the native Bluetooth stack of your platform, without a Sylvac dongle.

Choose your approach

Windows or Linux with a Sylvac USB dongle. Use the Sylvac libraries and examples of the developer repository. They handle scanning, connection, measurements and commands for you, and are supported by Sylvac.

Any platform with its own Bluetooth radio (smartphone, tablet, Mac, PC, embedded Linux). Talk to the instruments directly through the platform Bluetooth Low Energy API, following the Sylvac custom GATT profiles. The links below help you get started.

What you need to know about the instrument

- Sylvac instruments advertise a device name starting with `SY`. OEM compatible instruments use `MTY`.
- The measured value is provided by the **Simple Data Service (SDS)**.
- Commands and answers go through the **Sylvac Metrology Service (SMS)**, service `0x5000`: write the command on RemoteRequest `0x5012`, receive the answer on RemoteResponse `0x5013` (notify). DataSend `0x5010` (indicate) carries the value sent with the instrument button.
- Sylvac base UUID: `C1B2xxxx-CAAF-6D0E-4C33-7DAE30052840`
- SMS strings are plain ASCII terminated by a carriage return (`0x0D`), sent in chunks of 20 bytes at most.

Full details are in the *Sylvac custom Bluetooth GATT profiles* folder of the developer repository: github.com/SylvacDev/BluetoothDevTools

Windows

Windows.Devices.Bluetooth (WinRT, C#, C++)

- Overview: learn.microsoft.com/en-us/windows/uwp/devices-sensors/bluetooth-low-energy-overview
- API reference: learn.microsoft.com/en-us/uwp/api/windows.devices.bluetooth

Linux

BlueZ, the official Linux Bluetooth stack (D-Bus API)

- github.com/bluez/bluez/tree/master/doc

For most projects a library on top of BlueZ is easier, see *Cross-platform libraries* on the next page.

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Android

- Official guide: developer.android.com/develop/connectivity/bluetooth/ble/ble-overview
- Practical guide by Punch Through: punchthrough.com/android-ble-guide

iOS and macOS

Core Bluetooth (Swift, Objective-C)

- developer.apple.com/documentation/corebluetooth

Cross-platform libraries

Language / framework	Library	Link
Python	Bleak	github.com/hbldh/bleak
Rust	btleplug	github.com/deviceplug/btleplug
Node.js	@abandonware/noble	github.com/abandonware/noble
.NET MAUI	Plugin.BLE	github.com/dotnet-bluetooth-le/dotnet-bluetooth-le
Flutter	flutter_blue_plus	github.com/chipweinberger/flutter_blue_plus
React Native	react-native-ble-plx	github.com/dotintent/react-native-ble-plx
Ionic / Capacitor	@capacitor-community/bluetooth-le	github.com/capacitor-community/bluetooth-le

Support

Sylvac supports its libraries, its USB dongles and the Sylvac custom GATT profiles. The third-party APIs and libraries listed in this document are given as starting points and are not supported by Sylvac.